

Survey Me! — Using external data
comparisons to flag issues in OSM
<https://osm.mathmos.net/survey/>

Robert Whittaker

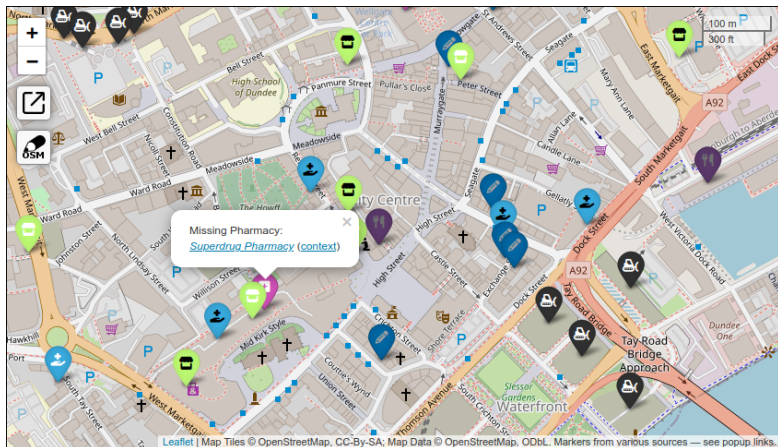
SotM Europe 2025

About Me

- OSM Mapper since 2008:
<https://www.openstreetmap.org/user/Robert%20Whittaker>
- Day job: Maths Lecturer at the University of East Anglia
<https://robert.mathmos.net/research/>
- Enjoy walking, cycling and running
- Interest in civic engagement, government transparency, and open data
- Author of various OSM tools:
<https://osm.mathmos.net/>
- OSM editor of choice: **JOSM**



The “Survey Me!” Tool (<https://osm.mathmos.net/survey/>)



- Combines output from other comparison and QA tools
- Flags UK issues where a ground survey would be useful

Purpose / Aims / Motivation

- Wanted a way to see results of my various OSM data comparison tools in one place.
- Wanted something to tell me nearby locations that would be worth a quick check when visiting somewhere new with non-OSM friends / family.
- Lots of other great mapping and QA apps available:
 - [StreetComplete](#)
 - [Every Door](#)
 - [Map Complete](#)
 - [OSM Inspector](#)
 - [Osmose](#)
- But these typically either
 - Also flag many things that can be fixed remotely
 - Flag too many issues to survey for my use-case

UK Data Categories in Survey Me!

External Data Comparisons

- Retail Chain Branches
(*AllThePlaces + RetailPoints*)
- Healthcare Locations
(*NHS Open Data*)
- Pharmacies
(*GPhC Register*)
- Public Defibrillators
(*The Circuit + NDDB*)
- Schools
(*Government Open Data*)
- Post Offices
(*FOI list from Post Office Ltd*)
- Postboxes
(*Old FOI list from Royal Mail*)

Internal OSM QA Checks

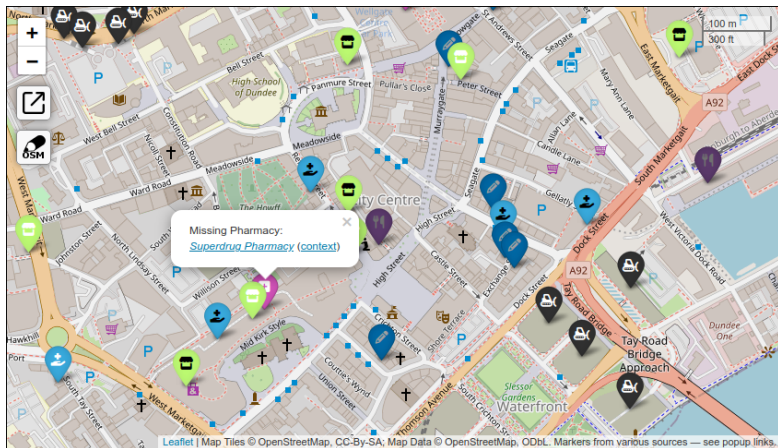
- Defunct retail chains
(*Overpass queries*)
- POIs without `name=*`
(*Overpass query*)
- Notes with `#surveyme`
(*OSM Notes API*)
- Stale Developments
(`construction=*` not edited in the last year)
(*Overpass query*)
- Objects with defunct Food Hygiene IDs
(`fhrs:id=*`) values
(*Feed from [FHODOT](#) tool*)

Survey Me! Numbers as of November 2025

External Dataset	Data Points	OSM Mapped	OSM Missing	OSM Un-matched
Pharmacies	12 491	9 170	4 130 (33%)	809 (9%)
Post Offices	14 516	9 347	6 092 (42%)	923 (10%)
Schools	29 693	30 854	1 342 (5%)	2 003 (6%)
Healthcare	49 327	25 012	30 711 (62%)	6 384 (26%)
Public Defibs	61 625	19 933	48 554 (79%)	2 428 (12%)
Retail Chains	104 030	80 991	32 448 (31%)	9 013 (11%)
Postboxes	115 485	90 319	28 046 (24%)	2 689 (3%)
Total	387 167	265 626	149 981 (39%)	24 249 (9%)

QA Check	# Issues
Defunct retail chains stores	716
OSM Notes with #surveyme	1 099
Stale developments	19 306
Defunct Food Hygiene IDs	35 552
Nameless POIs	52 732
Total	109 405

Survey Me! Map View

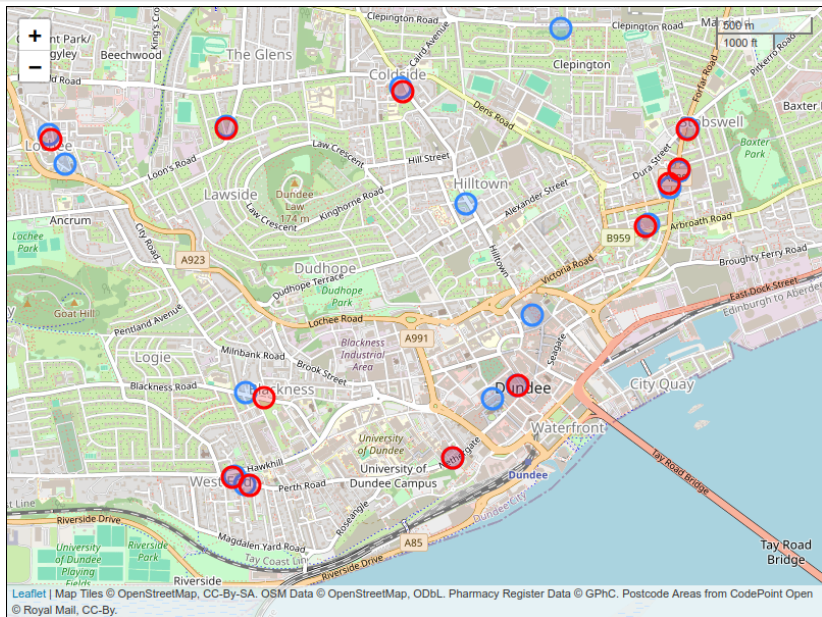


- Interactive map at <https://osm.mathmos.net/survey/>.
- GeoJSON, GPX & KML downloads available to load in GPS receivers and mapping apps.

Obtaining OSM and External Data

- Fetch OSM data using [Overpass API](#):
 - Script to perform update for each tool/category
 - Run manually roughly once a week
 - For larger datasets, split the UK into a grid of smaller rectangles and request each one in turn
 - Store object ID, lat, lon and other relevant tags.
- Obtain external data from:
 - Official open datasets
 - Data obtained under FOI
 - Scraping company websites
- Store everything in a MariaDB database on a shared web-server ([Mythic Beasts](#)).
- Scripting / matching done with (badly written) Perl.

External Dataset (red) + OSM Data (blue)



Data Matching / Comparison

Attempt to match up OSM objects with records in external data:

- 1 **Unique IDs** (*If available and compatibly-licensed*)
 - Match closest OSM object tagged with matching ID.
- 2 **Postcodes**
 - Match closest OSM object tagged with matching `addr:postcode=* tag`.
- 3 **Location Proximity**
 - Iterate through unmatched objects in one dataset.
 - Find all un-matched objects within a fixed distance d in the other dataset.
 - Match closest object.
 - Repeat for ~ 3 increasing distances d .

Remaining unmatched objects in both datasets are then flagged as errors / warnings / places to survey.

Can also flag other tagging issues in the OSM data.

What makes a good external dataset ?

- One-to-one correspondence with queriable OSM object set
- Completeness / accuracy
- Reasonably accurate location data
- Persistent unique IDs
- Regular updates
- Consistent data format over time
- Ability to fully automate download and import
- OSM-compatible licence (OGL / ODbL)
- Owner receptive to feedback / error reports

One-to-one Correspondence Failures

False positives arise from:

- Entries in the data that shouldn't be mapped in OSM.

Examples:

- Retail concessions in department stores may appear in a brand's store-finder, but wouldn't normally be mapped separately in OSM.
- Defibrillator data includes AEDs inside private office buildings, which won't often be mapped.

Potential work-arounds:

- Pre-process dataset to remove these entries.
- Keep entries but don't flag as errors if not mapped in OSM.

One-to-one Correspondence Failures

False positives arise from:

- Entries not in the data that are legitimately mapped in OSM with the same tagging.

Examples:

- Not all `amenity=post_office` objects in the UK are the Government-owned Post Office Ltd.
- `healthcare=clinic` used for both stand-alone clinics (in dataset) and hospital departments (not in dataset).

Potential work-arounds:

- Secondary tagging to distinguish OSM objects.

One-to-one Correspondence Failures

False positives arise from:

- Entries in the data that correspond to more than one legitimately mapped object in OSM.

Examples:

- A single GP Doctors' Practice (one data entry) can operate from multiple locations.
- Supermarkets with petrol stations will often have a `shop=convenience` on the forecourt in addition to the main `shop=supermarket`.

Potential work-arounds:

- Match via IDs, and allow multiple matches with same ID.
- Other tagging to identify sets of OSM objects that should be grouped together, or that aren't the primary object.

Use of AllThePlaces data for retail chains

- **AllThePlaces** (<https://alltheplaces.xyz/>) is an open-source project that scrapes company websites to obtain lists of branches / stores.
 - Provides standard datasets of branch locations, with addresses, opening hours, and other metadata.
 - Major source of external data for Survey Me!
 - ATP has slightly different objectives from the ideal for an OSM comparison tool — but that's fine.
 - My 'Chain Reaction' tool compares:
 - OSM objects with given `brand:wikidata=*` value
 - Branch list for that brand from AllThePlaces / other source.
- (See <https://osm.mathmos.net/chains/>.)

Pre-processing AllThePlaces data prior to import

- Check for broken scrapers, where there's lots of missing lots of branches. Replace these with previous good output.
- Filter out non-UK branches
 - e.g. Some chains list Irish branches with their UK ones.
- Filter out unwanted branch types
 - e.g. cash machines, supply kitchens, concessions
- Remove closed / duplicate branches
 - Some automatic, some manual
- Correct some assigned brands (to match OSM usage)
- Fix some names / addresses / websites
- Correct postcode errors

Improve OSM data to make comparison work

One-off checks prior to adding a new brand/chain, but periodic re-checks (once a year or so) also useful.

Check un-matched stores in dataset and OSM:

- Ensure all relevant OSM objects have `brand:wikidata=*` set correctly.
- Merge duplicate OSM objects.
- Split objects representing two different brands (e.g. post office and convenience store).
- Remove brand tags left behind on closed stores.

An **Overpass Turbo** query for OSM objects with a matching name without the right brand tag, can speed things up:

```
nwr ["name"="McDonald's"] ["brand:wikidata"!="Q38076"]
```

(Full query at <https://osm.mathmos.net/notes/overpass-queries.html>.)

Tricks in the tool to give better 1-to-1 matching

- Treat OSM objects with the same website as permitted duplicates.
 - e.g. Petrol station store attached to supermarket
- Ignore some OSM objects with matching `brand:wikidata`
 - `shop=outpost, amenity=atm, amenity=vending_machine, commercial=office|warehouse|distribution_centre, advertising=*` without other `shop=*, amenity=*` etc.
- Special treatment of some brands on the Channel Islands, which are run under franchise arrangements, and don't appear in the ATP data.
- A few other special cases, e.g. *WHSmith* sold its high street stores, but the re-branding to *TGJones* is still ongoing. So fetch both brands from OSM for now.
- Mechanism to suppress individual stores in the data.

Final Thoughts

- Lots of good data that can be used to help improve OSM.
- But licensing and data-quality / precision issues mean not suitable for direct imports. So run comparisons instead.
- Significant OSM cleanup and/or data pre-processing often needed to avoid lots of false positives.
- Survey Me! combines output from comparison and QA tools to flag issues that would benefit from a ground survey.
- Give it a try next time you visit somewhere new in the UK:
<https://osm.mathmos.net/survey/>.
- Try building something similar for your own country/region.
- To create a good tool, you need to understand (and work around) the limitations of both OSM and your external data.

For

- Contact details
- Survey Me!
- My other OSM Tools
- Slides from this talk

Visit my OSM website:

<https://osm.mathmos.net/>

